Pengcheng Lu

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Research Interests

Auditing AI judgments and behavior in domains without ground truth, from models to deployed systems and agents; construct validity of AI evaluation; culture as a testbed for how signals such as best-seller lists and platform communities enter language models; computational social science.

Education

University of Washington, Seattle, WA — Expected June 2027
M.S. in Information Management (Data Science) · GPA 3.91/4.0
Coursework: Data Visualization, Quantitative Research Methods, Data Science

Miami University, Oxford, OH — May 2025
B.S. in Commerce · GPA 3.8/4.0 · Graduated with Distinction · President's List

Research Experience

Auditing Language-Model Judgments of Cultural Works — Sep 2025 – present
University of Washington · with Taixin Zhang

- Ran a preregistered audit of how models from four families reproduce the Goodreads BookTok shelf: they returned a small, repeated set of top titles and left most of the shelf out (under review, ICWSM 2027).
- Designed controlled wording experiments on seven open-weight models: describing a book as among the "leading titles" of a best-seller list moved their judgments, while stating only its rank barely did.
- Built a claim-verification pipeline for 679 contemporary novels across four open-weight models, with an LLM judge calibrated against a human-labeled gold set; only 7.9–16.9% of the content details the models produced were supported by checked references.

Pre-Deployment Audit Tool for AI Agents — Sep 2026 – June 2027
MSIM Capstone, University of Washington · sponsored by Amazon · with Taixin Zhang

- Designing an open-source tool for Amazon Bedrock AgentCore that scans an agent, runs it in a sandbox, drafts a least-privilege permission policy and produces an evidence report for approvers; now in requirements interviews.

Best-Seller Lists and Public-Library Borrowing — Feb 2026 – present
Independent study in a cultural analytics project led by Professor Melanie Walsh, University of Washington

- Collected the New York Times best-seller lists from 2008 to 2026 through the NYT Books API.
- Built a Python (Polars) pipeline to clean about 20 million Seattle Public Library checkout records, resolved messy titles to works with an LLM-assisted normalization step, and linked them to list titles with constrained fuzzy matching.
- Analyzing how appearing on a best-seller list relates to public-library borrowing over time.

AI-Disclosure Labels and Ad Responses — Spring 2025
Undergraduate capstone research, Department of Commerce, Miami University · supervisor: Dr. Tom Mays

- Ran the study end to end: framed the question, prepared ten pairs of human-made and AI-generated ad posters, built a counterbalanced Qualtrics survey, collected responses from 21 participants and analyzed the data. Labeling ads as AI-generated produced no reliable average change in attention, interest, desire or action.
- Report co-written with Dr. Mays: "Visual Ads in the Age of AI: How Gen Z's Attitudes Shape Ad Effectiveness."

Papers

Under review

Pengcheng Lu and Taixin Zhang. "When BookTok Becomes a Prompt: How Language Models Reorder the Record of a Platform Reading Culture." ICWSM 2027. Preregistered at OSF.

Working papers

Pengcheng Lu and Taixin Zhang. "A Shelf and a List: How Two Records of Book Attention Fare When a Language Model Reads Them."

Pengcheng Lu and Taixin Zhang. "Verifying Cultural Information about Contemporary Fiction: Language-Model Claims and Public Textual Presence."

Teaching and Service

Reader/Grader, Information School, University of Washington — Summer 2026
IMT 550 B: Policy and Ethics in Information Management · instructor: Mark Dennis Robinson
Graded weekly work and wrote feedback for 23 students; handled student communication and accommodations with the instructor.

Reviewer, ICWSM 2027 (two papers) — 2026

Honors and Awards

Larsen Endowed Fellowship, Information School, University of Washington — 2026–2027
Alternate team (one of two among 67), UW Hollomon Health Innovation Challenge (MindBridge) — 2026
2nd place of 45 teams, Husky Hackathon, University of Washington (Sound Quest) — 2025
Graduated with Distinction; President's List, Miami University — 2025

Selected Projects

One IKEA Model, Two Regional Playbooks · interactive data article, with Samantha Wang — Spring 2026
Assembled six sourced datasets and built five visualizations and the scrolling page.

MindBridge · early-warning system for student mental-health support; led a five-person team — 2026

Sound Quest · gamified financial dashboard, Husky Hackathon; led a four-person team — 2025

Professional Experience

Huayu Electronics, Business Analyst, Chengdu, China — May – Aug 2024
Mapped production workflows through stakeholder interviews and process observation, built scenario models comparing in-house, automation and outsourcing options, and presented the recommendation to company leadership, who approved the automation investment.

China Unicom, Market Analysis Intern, Chengdu, China — May – Aug 2023
Segmented over 1 million user records with SQL and Python into behavioral personas and built a Power BI dashboard that supported a new data plan for students.

China Tower, Financial Analysis Intern, Chengdu, China — Mar – Aug 2022
Traced reconciliation errors in settlement records to inconsistent data standards and helped standardize fields across systems.

Skills

Research methods: experimental design (Qualtrics), preregistration, equivalence testing, construct validation, regression, entity resolution

AI evaluation: running and auditing open-weight language models, LLM APIs, LLM-as-judge calibration

Programming: Python (Polars, pandas, RapidFuzz), SQL, Stata

Visualization: p5.js, Tableau, Power BI, Matplotlib, Figma

Languages: Chinese (native), English